

Waves Motion

Single Correct Type Questions

1. Which of the following equations represents a traveling wave?
[24 Feb, 2021 (Shift-II)]

- (a) $y = A \sin x \cos \omega t$
(b) $y = A \sin(15x - 2t)$
(c) $y = Ae^x \cos(\omega t - \theta)$
(d) $y = Ae^{-x^2} (vt + \theta)$

2. A travelling harmonic wave is represented by the equation $y(x, t) = 10^{-3} \sin(50t + 2x)$, where x and y are in meter and t is in seconds. Which of the following is a correct statement about the wave?
[12 Jan, 2019 (Shift-I)]

- (a) The wave is propagating along the negative x -axis with speed 25 ms^{-1} .
(b) The wave is propagating along the positive x -axis with speed 100 ms^{-1} .
(c) The wave is propagating along the positive x -axis with speed 25 ms^{-1} .
(d) The wave is propagating along the negative x -axis with speed 100 ms^{-1} .

3. A steel wire with mass per unit length $7.0 \times 10^{-3} \text{ kg}$ is under tension of 70 N . The speed of transverse waves in the wire will be:
[01 Feb, 2023 (Shift-I)]

- (a) $200 \pi \text{ m/s}$ (b) 100 m/s
(c) 10 m/s (d) 50 m/s

4. A travelling wave is described by the equation

$$y(x, t) = [0.05 \sin(8x - 4t)] \text{ m}$$

The velocity of the wave is: [all the quantities are in SI unit]
[24 Jan, 2023 (Shift-I)]

- (a) 4 ms^{-1} (b) 2 ms^{-1}
(c) 0.5 ms^{-1} (d) 8 ms^{-1}

5. In the wave equation

$$y = 0.5 \sin \frac{2\pi}{\lambda} (400t - x) \text{ m}$$

the velocity of the wave will be: [28 July, 2022 (Shift-I)]

- (a) 200 m/s
(b) $200\sqrt{2} \text{ m/s}$
(c) 400 m/s
(d) $400\sqrt{2} \text{ m/s}$

6. Equation of travelling wave on a stretched string of linear density 5 g/m is $y = 0.03 \sin(450t - 9x)$ where distance and time are measured in SI units. The tension in the string is:
[11 Jan, 2019 (Shift-I)]

- (a) 10 N (b) 7.5 N (c) 12.5 N (d) 5 N

7. The equations of two waves are given by:

$$y_1 = 5 \sin 2\pi(x - vt) \text{ cm}$$

$$y_2 = 3 \sin 2\pi(x - vt + 1.5) \text{ cm}$$

These waves are simultaneously passing through a string. The amplitude of the resulting wave is:

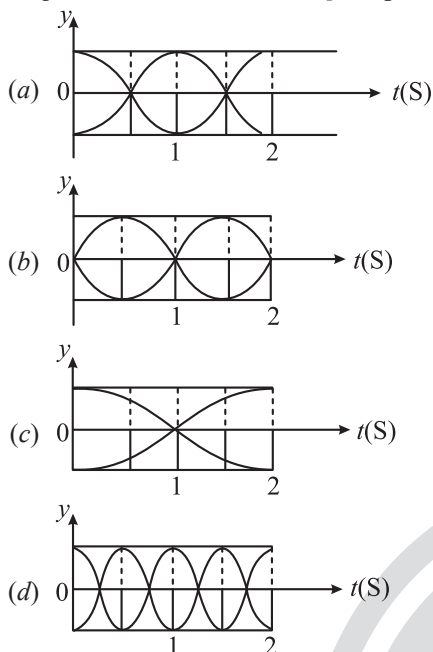
[24 June, 2022 (Shift-I)]

- (a) 2 cm (b) 4 cm
(c) 5.8 cm (d) 8 cm

8. Three harmonic waves having equal frequency ν and same intensity I_0 , have phase angles 0 , $\frac{\pi}{4}$ and $-\frac{\pi}{4}$ respectively. When they are superimposed the intensity of the resultant wave is close to
[9 Jan, 2020 (Shift-I)]

- (a) $5.8 I_0$ (b) $3 I_0$
(c) I_0 (d) $0.2 I_0$

9. The correct figure that shows, schematically, the wave pattern produced by superposition of two waves of frequencies 9 Hz and 11 Hz is: [10 April, 2019 (Shift-II)]



10. Two coherent sources produce waves of different intensities which interfere. After interference, the ratio of the maximum intensity to the minimum intensity is 16. The intensity of the waves are in the ratio

[9 Jan, 2019 (Shift-I)]

- (a) 16 : 9 (b) 25 : 9
(c) 4 : 1 (d) 5 : 3

11. Two identical strings X and Z made of same material have tension T_x and T_z in then if their fundamental frequencies are 450 Hz and 300 Hz, respectively, then the ratio T_x/T_z is:

[2 Sep, 2020 (Shift-I)]

- (a) 1.25 (b) 2.25
(c) 1.5 (d) 0.44

12. A wire of length L and mass per unit length $6.0 \times 10^{-3} \text{ kg m}^{-1}$ is put under tension of 540 N. Two consecutive frequencies that it resonates at are 420 Hz and 490 Hz. Then L in meters is:

[9 Jan, 2020 (Shift-II)]

- (a) 1.1 m (b) 5.1 m
(c) 2.1 m (d) 8.1 m

13. A wire of length $2L$, is made by joining two wires A and B of same lengths but different radii r and $2r$ and made of the same material. It is vibrating at a frequency such that the joint of the two wires forms a node. If the number of anti-nodes in wire A is p and that in B is q then the ratio $p : q$ is:

[8 April, 2019 (Shift-I)]



- (a) 4 : 9 (b) 3 : 5
(c) 1 : 4 (d) 1 : 2

14. A string is clamped at both the ends and it is vibrating in its 4th harmonic. The equation of the stationary wave is $Y = 0.3 \sin(0.157x) \cos(200\pi t)$. The length of the string is: (All quantities are in SI units.)

[9 April, 2019 (Shift-I)]

- (a) 20 m (b) 80 m
(c) 60 m (d) 40 m

15. The ratio of speed of sound in hydrogen gas to the speed of sound in oxygen gas at the same temperature is:

[06 April, 2023 (Shift-II)]

- (a) 4 : 1 (b) 1 : 2 (c) 1 : 4 (d) 1 : 1

16. In a resonance tube experiment when the tube is filled with water up to a height of 17.0 cm from bottom, it resonates with a given tuning fork. When the water level is raised the next resonance with the same tuning fork occurs at a height of 24.5 cm. If the velocity of sound in air is 330 m/s, the tuning fork frequency is

[05 Sep, 2020 (Shift-I)]

- (a) 2200 Hz (b) 3300 Hz
(c) 1100 Hz (d) 550 Hz

17. A closed organ pipe has a fundamental frequency of 1.5 kHz. The number of overtones that can be distinctly heard by a person with this organ pipe will be: (Assume that the highest frequency a person can hear is 20,000 Hz)

[10 Jan, 2019 (Shift-II)]

- (a) 6 (b) 4
(c) 7 (d) 5

Integer Type Questions

18. The amplitude of wave disturbance propagating in the positive x -direction is given by $y = \frac{1}{1+(x)^2}$ at $t = 0$ and $y = \frac{1}{1+(x-2)^2}$ at $t = 1$ s, where x and y are in metres. The shape of wave does not change during the propagation. The velocity of the wave will be _____ m/s.

[20 July, 2021 (Shift-I)]

19. A guitar string of length 90 cm vibrates with a fundamental frequency of 120 Hz. The length of the string producing a fundamental frequency of 180 Hz will be _____ cm.

[08 April, 2023 (Shift-II)]

20. In an experiment with sonometer when a mass of 180 g is attached to the string, it vibrates with fundamental frequency of 30 Hz. When a mass m is attached, the string vibrates with fundamental frequency of 50 Hz. The value of m is _____ g.

[13 April, 2023 (Shift-II)]

21. A wire of length 30 cm, stretched between rigid supports, has its n^{th} and $(n + 1)^{\text{th}}$ harmonics at 400 Hz and 450 Hz, respectively. If tension in the string is 2700 N, its linear mass density is _____ kg/m. [27 July, 2022 (Shift-II)]
22. The percentage increase in the speed of transverse waves produced in a stretched string if the tension is increased by 4%, will be%. [25 Feb, 2021 (Shift-II)]
23. A wire having a linear mass density 9.0×10^{-4} kg/m is stretched between two rigid supports with a tension of 900 N. The wire resonates at a frequency of 500 Hz. The next higher frequency at which the same wire resonates is 550 Hz. The length of the wire is _____ m. [31 Aug, 2021 (Shift-I)]
24. Two travelling waves produces a standing wave represented by equation, $y = 1.0 \text{ mm} \cos(1.57 \text{ cm}^{-1})x \sin(78.5 \text{ s}^{-1})t$. The node closest to the origin in the region $x > 0$ will be at $x =$ _____ cm. [26 Aug, 2021 (Shift-I)]
25. For a certain organ pipe, the first three resonance frequencies are in the ratio of 1 : 3 : 5 respectively. If the frequency of fifth harmonic is 405 Hz and the speed of sound in air is 324 ms^{-1} the length of the organ pipe is _____ m. [12 April, 2023 (Shift-I)]
26. In an experiment to determine the velocity of sound in air at room temperature using a resonance tube, the first resonance is observed when the air column has a length of 20.0 cm for a tuning fork of frequency 400 Hz is used. The velocity of the sound at room temperature is 336 ms^{-1} . The third resonance is observed when the air column has a length of _____ cm. [29 June, 2022 (Shift-II)]
27. A tuning fork of frequency 340 Hz resonates in the fundamental mode with a air column of length 125 cm in a cylindrical tube closed at one end. When water is slowly poured in it, the minimum height of water required for observing resonance once again is _____ cm. (Velocity of sound in air is 340 ms^{-1}) [28 June, 2022 (Shift-II)]
28. A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with same frequency. The length of the open pipe is $\frac{x}{3}L\sqrt{\frac{\rho_1}{\rho_2}}$ where x is (Round off to the Nearest Integer) [16 March, 2021 (Shift-II)]
29. A stationary tuning fork is in resonance with an air column in a pipe. If the tuning fork is moved with a speed of 2 ms^{-1} in front of the open end of the pipe and parallel to it, the length of the pipe should be changed for the resonance to occur with the moving tuning fork. If the speed of sound in air is 320 ms^{-1} , the smallest value of the percentage change required in the length of the pipe is _____. [JEE Adv, 2020]

ANSWER KEY

1. (b) 2. (a) 3. (b) 4. (c) 5. (c) 6. (c) 7. (a) 8. (a) 9. (d) 10. (b)
 11. (b) 12. (c) 13. (d) 14. (b) 15. (a) 16. (a) 17. (c) 18. [1] 19. [60]
 20. [500] 21. [3] 22. [2] 23. [10] 24. [1] 25. [1] 26. [104] 27. [50] 28. [4]
 29. [0.62 to 0.63]

EXPLANATIONS

1. (b) We know that, travelling wave can be represented as

$$f\left(t \pm \frac{x}{v}\right)$$

Thus, $y = A \sin(15x - 2t)$ represents a travelling wave.

2. (a) $50t + 2x = \text{constant}$

$$50 + \frac{2dx}{dt} = 0$$

$$v = \frac{dx}{dt} = -25 \text{ m/s}$$

3. (b) $\therefore v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{70}{7 \times 10^{-3}}} = 100 \text{ m/s}$

4. (c) From the given equation $k = 8 \text{ m}^{-1}$ and $\omega = 4 \text{ rad/s}$

$$\text{Velocity of wave} = \frac{\omega}{k}$$

$$v = \frac{4}{8} = 0.5 \text{ m/s}$$

5. (c) We have, $y = 0.5 \sin \frac{2\pi}{\lambda} (400t - x) \text{ m}$

$$\text{So, } \omega = \frac{2\pi}{\lambda} \times 400 \text{ and } K = \frac{2\pi}{\lambda}$$

$$\therefore \text{Wave speed, } V = \frac{\omega}{k} = 400 \text{ m/s}$$

6. (c) $v = \frac{\omega}{k} = \frac{450}{9} = 50 \text{ ms}^{-1}$

$$v = \sqrt{\frac{T}{\mu}}, T = v^2 \mu = 50^2 \times 5 \times 10^{-3} = 12.5 \text{ N}$$

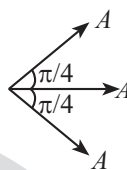
7. (a) $y_1 = 5 \sin 2\pi(x - vt) \text{ cm}$, $y_2 = 3 \sin 2\pi(x - vt + 1.5) \text{ cm}$
 phase angle, $\phi = 2\pi \times 1.5 = 3\pi$

$$\therefore A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

$$= \sqrt{5^2 + 3^2 + 2 \times 5 \times 3 \cos 3\pi}$$

$$= \sqrt{25 + 9 + (30 \times -1)} = \sqrt{4} = 2 \text{ cm}$$

8. (a)



$$A_{\text{res}} = (\sqrt{2} + 1)A$$

$$I_{\text{res}} = (\sqrt{2} + 1)^2 I_0 = (3 + 2\sqrt{2}) I_0 = 5.8 I_{0s}$$

9. (d) $f_{\text{beat}} = 11 - 9 = 2 \text{ Hz}$

$\therefore$ Time period of oscillation of amplitude =

$$\frac{1}{f_{\text{beat}}} = \frac{1}{2}$$

Although the graph of oscillation is not given, the equation of envelope is given by option (d)

10. (b) $\left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right) = \frac{16}{1}; \frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} = \frac{4}{1}$

$$\Rightarrow 3\sqrt{I_1} = 5\sqrt{I_2}$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{25}{9}$$

11. (b) We know that,

$$f_x = \frac{1}{2\ell} \sqrt{\frac{T_x}{\mu}}; f_z = \frac{1}{2\ell} \sqrt{\frac{T_z}{\mu}}$$

$$\text{Here, } f \propto \sqrt{T}$$

$$\frac{f_x}{f_z} = \frac{450}{300} = \sqrt{\frac{T_x}{T_z}} \Rightarrow T_x/T_z = 9/4 = 2.25$$

Hence, option (B) is correct answer

12. (c) Fundamental frequency = $490 - 420 = 70 \text{ Hz}$

$$70 = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

($\therefore \mu$ is mass per unit length and T is tension in the wire)

$$\Rightarrow 70 = \frac{1}{2\ell} \sqrt{\frac{540}{6 \times 10^{-3}}}$$

$$\Rightarrow \ell = \frac{1}{2 \times 70} \sqrt{90 \times 10^3} = \frac{300}{140} = 2.14 \text{ m}$$

$$13. (d) \therefore \mu_1 = \frac{\rho \pi r^2 L}{L} = \mu \text{ and } \mu_2 = \frac{\rho 4 \pi r^2 L}{L} = 4\mu$$

Tension in both are the same = T

Let the speed of the wave in the wires be V_1 and V_2

$$V_1 = \sqrt{\frac{T}{\mu}} = V, V_2 = \sqrt{\frac{T}{4\mu}} = \frac{V}{2}$$

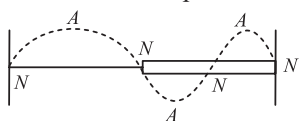
So fundamental frequencies in both wires are

$$f_1 = \frac{V_1}{2L} = \frac{V}{2L} \text{ and } f_2 = \frac{V_2}{2L} = \frac{V}{4L}$$

Frequency at which both resonate is L.C.M. of both

frequencies i.e., $\frac{V}{2L}$.

Hence no. of loops in wires are 1 and 2 respectively.



So, ratio of no. of antinodes is 1 : 2.

$$14. (b) \text{ 4th harmonic } \left[\text{Diagram of a string fixed at both ends showing 4 loops (antinodes) and 5 nodes (including ends). The length is labeled } \ell \text{.} \right]$$

$$4 \frac{\lambda}{2} = \ell$$

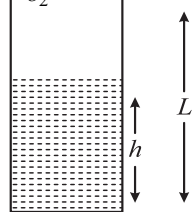
$$2\lambda = \ell$$

$$\frac{2\pi}{\lambda} = 0.157 \Rightarrow \lambda = 40$$

$$\therefore \ell = 2\lambda = 80 \text{ m}$$

$$15. (a) v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \frac{1}{\sqrt{M}}$$

$$16. (a) \frac{v_{H_2}}{v_{O_2}} = \sqrt{\frac{32}{2}} = \sqrt{\frac{16}{1}} = \frac{4}{1}$$



$$L - h_1 = n \frac{\lambda}{2} + \frac{\lambda}{4}$$

$$L - h_2 = (n-1) \frac{\lambda}{2} + \frac{\lambda}{4}$$

$$\Rightarrow \frac{\lambda}{2} = h_2 - h_1 = 24.5 - 17.0 = 7.5 \text{ cm}$$

$$\Rightarrow \lambda = 15 \text{ cm}$$

$$u = f\lambda$$

$$f = 2200 \text{ Hz}$$

$$17. (c) \text{ Fundamental frequency} = 1.5 \text{ kHz}$$

$$\text{Highest frequency} = 20000 \text{ Hz} = 20 \text{ kHz}$$

$$\text{Now, } \frac{20 \text{ kHz}}{1.5 \text{ kHz}} = 13.33$$

$\therefore$ Possible harmonics are 1, 3, 5, 7, 9, 11, 13 i.e., 7

$$18. [1] \text{ The equation of wave at any time } t \text{ can be written as,}$$

$$y = \frac{1}{1 + (x - vt)^2}$$

$$\text{Comparing above equation, } y = \frac{1}{1 + (x - vt)^2},$$

$$\text{we get } v = 1 \text{ m/s}$$

$$19. [60] \text{ Fundamental frequency, } f = \frac{v}{2\ell}$$

$$f \propto \frac{1}{\ell}$$

$$\frac{f_1}{f_2} = \frac{\ell_2}{\ell_1} \Rightarrow \ell_2 = \frac{f_1}{f_2} \times \ell_1$$

$$\Rightarrow \ell_2 = \frac{120}{180} \times 90 = 60 \text{ cm}$$

$$\ell_2 = 60 \text{ cm}$$

$$20. [500] \text{ Frequency of the string } f = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}} \text{ (T : Tension)}$$

$$\frac{f_2}{f_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{mg}{180g}}$$

$$\left(\frac{50}{30}\right)^2 = \frac{m}{180} \Rightarrow m = \frac{25}{9} \times 180 = 500 \text{ gram}$$

$$21. [3] \text{ For } n^{\text{th}} \text{ harmonic, } \frac{nv}{0.6} = 400$$

$$\text{For } (n+1)^{\text{th}} \text{ harmonic } \frac{(n+1)v}{0.6} = 450$$

$$\Rightarrow \left[\frac{0.6 \times 450}{v} + 1 \right] \frac{v}{0.6} = 450$$

$$\Rightarrow v = 30$$

$$\Rightarrow v = \sqrt{\frac{T}{\mu}} \Rightarrow v^2 = \frac{T}{\mu}$$

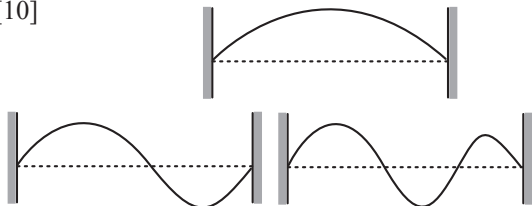
$$\Rightarrow \frac{2700}{\mu} = 900$$

$$\mu = 3$$

$$22. [2] v = \sqrt{\frac{T}{\mu}}, v' = \sqrt{\frac{1.04T}{\mu}}$$

$$\% \text{ change} = (\sqrt{1.04} - 1) \times 100 = 2\%$$

23. [10]



$$\therefore \ell = n \frac{\lambda}{2} \text{ (for wire fixed at both ends)}$$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{900}{9 \times 10^{-4}}} = 1000 \text{ m/s}$$

$$\therefore v = f\lambda \text{ or } \frac{v}{\lambda} = f \Rightarrow f_0 = \frac{nv}{2\ell}$$

$$f' = \frac{(n+1)v}{2\ell}$$

$$\frac{500 \text{ Hz}}{550 \text{ Hz}} = \frac{\frac{n \times 1000}{2\ell}}{\frac{(n+1) \times 1000}{2\ell}} \Rightarrow n = 10$$

$$\therefore 500 \text{ Hz} = \frac{10 \times 1000}{2 \times \ell}$$

$$\Rightarrow \ell = 10 \text{ m}$$

Hence, length of the wire is 10 m.

24. [1] From given equation, $K = 1.57 \text{ cm}^{-1}$

$$kx = \pi/2$$

$$\Rightarrow 1.57x = \frac{\pi}{2} = \frac{3.14}{2} = 1.57$$

$$x = 1 \text{ cm}$$

25. [1] Frequency of closed organ pipe $f_n = \frac{nv}{4\ell}$

For 5th harmonic,

$$f_5 = \frac{5V}{4} \Rightarrow 405 = \frac{5 \times 324}{4}$$

$$\Rightarrow \ell = 1 \text{ m}$$

$$26. [104] 400 = \frac{v}{4(L_1 + e)} \quad \dots(i)$$

$$400 = \frac{5v}{4(L_2 + e)} \quad \dots(ii)$$

$$\Rightarrow L_1 + e = \frac{\lambda}{4} = 21 \text{ cm}$$

$$\Rightarrow L_2 + e = \frac{5\lambda}{4} = 105 \text{ cm}$$

$$\Rightarrow e = 1 \text{ cm} \text{ \& } L_2 = 104 \text{ cm}$$

27. [50] We know that, $\lambda = \frac{v}{f} = \frac{340}{340} = 1 \text{ m}$

$$\text{First resonating length, } \lambda_1 = \frac{\lambda}{4} = 25 \text{ cm}$$

$$\text{Second resonating length, } \lambda_2 = \frac{3\lambda}{4} = 75 \text{ cm}$$

$$\text{Third resonating length, } \lambda_3 = \frac{5\lambda}{4} = 125 \text{ cm}$$

$$\text{Height of water required} = \lambda_3 - \lambda_2 = 125 - 75 = 50 \text{ cm}$$

28. [4] Both piper vibrate in first overtone.

$$f_c = f_o \Rightarrow \frac{3V_c}{4L} = \frac{2V_o}{2\ell_o} \Rightarrow \frac{3V_c}{4L} = \frac{V_o}{\ell_o}$$

$$\therefore \frac{V_o}{V_c} = \frac{3\ell_o}{4L}$$

$$\text{Also, } \frac{V_o}{V_c} = \sqrt{\frac{\rho_1}{\rho_2}}$$

$$\text{Then, } \sqrt{\frac{\rho_1}{\rho_2}} = \frac{3\ell_o}{4L} \Rightarrow \ell_o = \frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$$

29. [0.62 to 0.63]

$$f \propto \frac{1}{\ell_1} \Rightarrow f = \frac{k}{\ell_1} \quad \dots(i)$$

($\ell_1 \Rightarrow$ initial length of pipe)

$$\left(\frac{V}{V - V_T} \right) f = \frac{k}{\ell_2} \quad \dots(ii)$$

{ V_T speed of tuning fork, $\ell_1 \rightarrow$ new length of pipe}

(i) $\div$ (ii)

$$\frac{V - V_T}{V} = \frac{\ell_2}{\ell_1}$$

$$\frac{\ell_2}{\ell_1} - 1 = \frac{V - V_T}{V} - 1$$

$$\frac{\ell_2 - \ell_1}{\ell_1} = \frac{V_T}{V}$$

$$\frac{\ell_2 - \ell_1}{\ell_1} \times 100 = \frac{-2}{320} \times 100 = -0.625$$